Partial loss of eu-west storefront during CDN cutover

Postmortem for INC-5230 · 42 minutes · SEV-2 · author T. Iversen (Network Engineering) · reviewed 2026-08-11

Jira issue	INC-5230 · actions NET-1441 to NET-1445
Severity / status	SEV-2 · Closed 2026-08-12
Window	2026-08-06 02:14 – 02:56 UTC (42 minutes)
Blast radius	eu-west-1 storefront only; 18% of global traffic
Environment / service	Production · edge-cdn, storefront-web, ns1/ns2.helvetica.internal
Commander / comms	T. Iversen · L. Reber

Request path at time of failure

Figure 2 — only the cutover region is affected.

1. Description

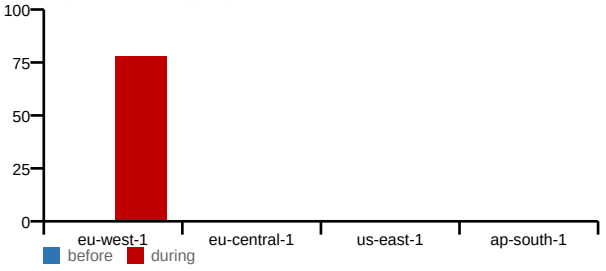
During a planned migration of the European storefront from the legacy CDN to edge-cdn, the DNS records for `www.helvetica-retail.eu` were repointed at 02:14 UTC on 6 August 2026. The new CNAME target was published with a trailing-dot error in the zone file, producing the fully qualified name `eu.edge-cdn.net.helvetica.internal` instead of `eu.edge-cdn.net`. That name does not exist, so every resolver that picked up the new record answered NXDOMAIN.

Customers in eu-west-1 whose resolver cache had expired could not reach the storefront at all: the browser reported a DNS failure rather than an error page, so the failure was invisible to our own HTTP monitoring. Customers with a warm cache were unaffected until their TTL expired, which is why the impact ramped up over the first fifteen minutes rather than starting at 100%.

2. Impact

- 42 minutes of partial unavailability, ramping from 0% to an estimated 78% of eu-west-1 sessions.
- Approximately 34,000 sessions unable to resolve the storefront.
- CHF 96,400 of estimated deferred revenue; CHF 11,300 assessed as lost.
- No impact to eu-central, us-east or ap-south, which were not part of this cutover wave.
- No data loss, no security impact.

Failed page loads, % by region



3. Timeline (UTC)

Time	Event
01:50	Change NET-CHG-8871 opened. Pre-checks pass.
02:14	Zone file applied to ns1/ns2. TTL is 300s.
02:16	Automated post-change check passes — it queries the storefront through the corporate resolver, whose cache is still warm.
02:21	First customer report via the eu support queue.
02:29	Second and third reports. L. Reber escalates to the change engineer.
02:31	T. Iversen declares SEV-2. Synthetic checks from an external vantage point confirm NXDOMAIN.
02:38	Zone file diff reviewed on the bridge; the trailing dot is spotted.
02:44	Corrected zone published; serial incremented.
02:49	Resolvers begin returning the correct answer as the 300s TTL expires.
02:56	External checks green from all five vantage points. Mitigated.
03:30	Cutover re-attempted successfully with the corrected record.
2026-08-11	Postmortem review held.

4. Root cause

A missing trailing dot in a hand-edited BIND zone file caused the CNAME target to be interpreted as relative to the zone origin. `eu.edge-cdn.net` without a terminating dot is expanded to `eu.edge-cdn.net.helvetica.internal`, a name that has never existed. The zone file passed `named-checkzone`, which validates syntax and cannot know that the operator meant an absolute name.

The error survived to production because the post-change verification queried the internal corporate resolver, which still held the previous answer in cache. A check from outside the network would have failed immediately.

5. Contributing factors

- Zone files are edited by hand and applied directly; there is no diff review step for DNS changes, unlike every other production

change.

- The change was scheduled at 02:14 local, when a single engineer was on duty and no reviewer was available.
- HTTP monitoring cannot see a DNS failure, and we had no DNS-resolution monitoring from outside the corporate network.

6. Action plan

Jira	Action	Owner	Due	Status
NET-1441	Move all production zones into version control with mandatory peer review before apply.	T. Iversen	2026-09-04	Done
NET-1442	Add a CI check that rejects any CNAME or MX target which is not fully qualified with a trailing dot.	T. Iversen	2026-08-21	Done
NET-1443	Add external DNS resolution probes from five public vantage points, alerting on NXDOMAIN or a changed answer.	A. Novak	2026-08-28	Done
NET-1444	Post-change verification must run from an external vantage point with a cold cache; update the change template accordingly.	L. Reber	2026-08-19	Done
NET-1445	Lower TTL to 60s for the 24 hours preceding any planned DNS cutover, and restore it afterwards.	T. Iversen	2026-08-26	Done

What went well. Once escalated, the diagnosis took seven minutes and the fix was a two-line change. The rollback plan for the cutover was written and tested in advance, and was ready to use had the corrected record not worked.